WILDA PROJECT | TEAM 4

Customer Churn Analysis

for a Telecommunications Company

Final Report: Stage 3, Predictive Modelling and Reporting

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Project period: 27 July to 25 October 2026

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Document Information and Contents

Item	Detail
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Submitted by	Bhawana K C, Project Manager (one submission for the whole team)
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Contents

Executive Summary	3
1. Introduction	4
2. Data and Methodology	6
Part A: Summary of Key Findings	9
Part B: Factors Contributing to Churn and Retention	12
Part C: Recommendations for Targeted Retention Strategies	14
Part D: Limitations and Proposed Solutions	17
3. Video Demonstration	18
4. Alignment with Stage 3 Requirements	19
5. Conclusion	19
Appendices	20

Executive Summary

A telecommunications company is losing customers faster than it can replace them. This project analysed the company's customer data to find out who leaves, why they leave, and what the business can do about it. The team prepared the data, grouped customers into behavioural segments with K-Means clustering, built an Artificial Neural Network (ANN) to predict which customers are likely to churn, and translated the results into retention actions.

The central finding is that churn is concentrated, not evenly spread. Of 7,043 customers, 1,869 (26.54%) have churned. One segment, new high spenders, makes up about 31.5% of the training customers but produces 57.8% of the training-set churners, and 61% of actual churners on the unseen test set. These customers have short tenure (14.9 months on average), pay high bills ($81.10 a month), and churn at roughly one in two. At the other end, long-term low spenders churn at only about 5%.

The neural network is a useful early-warning tool, with clear limits. On the 1,409-customer test set it identified 73.8% of real churners (recall), with an ROC-AUC of 0.808. Its accuracy was 72.3% and its precision for churners was 48.6%, so about half the customers it flags would have stayed. Permutation importance shows the model relies mainly on tenure (0.21) and monthly charges (0.11); other features contribute very little.

We recommend three segment-specific actions: (1) convert new high spenders to fixed-term contracts, (2) protect long-term high spenders with service and loyalty benefits rather than discounts, and (3) redirect spend away from long-term low spenders and towards upselling. New low spenders are monitored, not targeted at this stage.

The report is honest about what was not achieved. The project charter set targets of 85% accuracy and 80% recall. The model reached 72.3% and 73.8%, so these targets were not met. Part D explains why, and sets out concrete steps to close the gap.

Table 1: Headline results at a glance

Headline figure	Value	Source
Customers analysed	7,043	Data preparation (Rohan)
Customers who churned	1,869 (26.54%)	1,495 training + 374 test
Customer segments	4	K-Means, elbow method (Ramesh)
Highest-risk segment	New high spenders: 48.7% churn (training), 50.6% (test)	Clustering and ANN analysis
Share of churners from that segment	57.8% (training), 61% (test)	Clustering and ANN analysis
ANN recall / precision (churners)	73.8% / 48.6%	Test set (Shradha)
ANN accuracy / ROC-AUC	72.3% / 0.808	Test set (Shradha)
Strongest churn driver	Tenure (importance 0.21), then monthly charges (0.11)	Permutation importance

1. Introduction

1.1 Background

The telecommunications industry is changing quickly as digital technologies advance, yet one problem persists for leading providers: high customer churn. Winning a new customer typically costs far more than keeping an existing one, so understanding why customers leave, and acting before they do, protects both revenue and growth.

This project was delivered by a five-person team over 13 weeks in three stages. Stage 1 set up the project and produced the Project Charter. Stage 2 prepared the data and segmented customers. Stage 3, covered by this report, built and evaluated the predictive model and turned the findings into recommendations.

1.2 Problem Statement

The company cannot currently tell which customers are about to leave, or why. Its retention approach is reactive: it responds to cancellations after they happen. The strategic goal of this project is to move to a proactive approach, identifying and retaining at-risk customers before they churn.

1.3 Objectives and Outcome

The Project Charter set three SMART objectives. The table below states plainly how each one turned out.

Table 2: Charter objectives and outcomes

Objective (from charter)	Target	Outcome	Status
1. Identify and rank key churn drivers	Ranked list of the top 5 predictors with statistical evidence	Tenure (0.21) and monthly charges (0.11) clearly dominate; remaining features contribute very little, so a meaningful ranking beyond the top two is not supported by the data	Partly met
2. Develop a high-performance predictive model	Accuracy of at least 85% and recall of at least 80% on the test set	Accuracy 72.3%, recall 73.8%, ROC-AUC 0.808	Not met
3. Translate insights into retention strategies	At least three data-backed recommendations, each with a target segment and a specific action	Three prioritised recommendations plus one monitoring recommendation (Part C)	Met

1.4 Scope

Table 3: Project scope summary

In scope	Out of scope
Data preparation and exploratory analysis	Changes to IT systems, software, hardware or networks
Customer segmentation with K-Means clustering	Running any retention campaign, pricing change or operational change (we recommend, we do not implement)
ANN model to predict churn, with evaluation	Collecting new data through surveys, interviews or external sources
Identifying the factors behind churn and retention	Deploying the model to a production environment
At least three data-backed retention recommendations	Monitoring the effect of strategies after implementation

In scope	Out of scope
Final report, video and stakeholder presentation	Detailed financial modelling of retention strategies

1.5 Team and Responsibilities

Table 4: Team roles and contributions

Team member	Role	Contribution to this project
Bhawana K C	Project Manager	Chaired meetings, maintained Jira and Confluence, managed the repository and submissions, tracked the plan
Rohan Sharma Kharel	Data Engineer	Cleaned, encoded, split and scaled the data; found and fixed a scaling leakage problem; checked model inputs
Ramesh Kunwar	Data Analyst, Clustering	Chose the number of clusters, trained K-Means, produced and labelled the four segments
Shradha Malla	Data Analyst, Predictive Modelling; Deputy PM	Designed, trained and evaluated the ANN; produced predictions and feature importance
Himal Kunwar	Business Analyst	Churn driver analysis, key findings, retention recommendations, limitations, and the consolidated report

1.6 Project Stages and Deliverables

Table 5: Project stages

Stage	Focus	Main deliverables
Stage 1	Planning and setup	Project Charter: scope, goals, risks, timeline
Stage 2	Data preparation and modelling	Preprocessed dataset, train and test sets, scaling; elbow method, trained K-Means model, labelled cluster visualisations
Stage 3	Prediction and reporting	ANN architecture, trained model and evaluation; this Final Report; video demonstration

2. Data and Methodology

The analysis ran as one pipeline in which each stage fed the next: data preparation, customer segmentation, then predictive modelling. This section summarises each stage from a business point of view. Code, notebooks and technical documents are in the project repository.

2.1 The Dataset

Table 6: Dataset overview

Measure	Value
Customers	7,043
Columns in the raw dataset	10
Model-ready files	10 input features plus the churn target (11 columns)
Missing values	None found; handling step kept in the pipeline for future data
Duplicate rows	302, kept (no customer ID, so identical rows can be different customers)
Churn rate	26.54% overall; 73.5% of customers stayed and 26.5% left
Training set	5,634 customers (1,495 churners)
Test set	1,409 customers (374 churners)

The two classes are imbalanced: about three customers stay for every one who leaves. This shaped how the model was trained and how its results are judged, because a model that simply predicted “no churn” for everyone would already be right 73.5% of the time.

2.2 Data Preparation

- **Quality checks.** The data was checked for missing values, duplicates and inconsistent entries. Median fills numeric gaps and the mode fills categorical gaps, although none were needed.
- **Encoding.** Two-value columns were label-encoded to 0 and 1. Contract has three values, so it was one-hot encoded; numbering it 0, 1 and 2 would wrongly tell a model that a two-year contract is twice a one-year contract.
- **Stratified split.** The data was split 80/20 into training and test sets, stratified on churn so each holds exactly 26.54% churners. A fixed random state (42) makes the split repeatable.
- **Scaling.** Tenure (0 to 72 months) and monthly charges ($18 to $118) were standardised so that dollar amounts do not dominate distance calculations just because their numbers are larger.
- **Leakage control.** The scaler was fitted on the training set only and then applied to the test set. Fitting on everything would let test data influence training and make results look better than they are. This problem was found and fixed before submission.

2.3 Customer Segmentation

Customers were grouped with K-Means clustering on the 5,634 scaled training rows, using 10 features. The churn label was deliberately left out, so the segments describe behaviour and are not a restatement of the answer; churn was used afterwards only to measure each finished segment.

The number of clusters came from the elbow method, testing every k from 1 to 10. Each extra cluster lowers the total squared distance (inertia), but by less and less. The fall was 25.2% at two clusters, 18.0% at three, 12.1% at four and only 5.1% at five. Four is the last cluster that buys a real improvement.

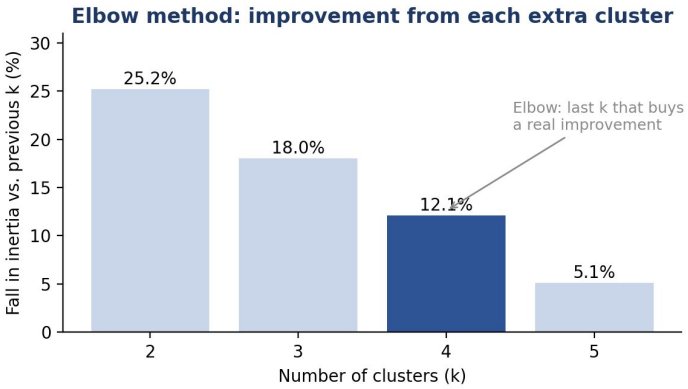


Figure 1: Elbow method: fall in inertia from each additional cluster (k = 4 selected)

The silhouette score peaks at two clusters (0.227) and is 0.203 at four. Four was still chosen: two clusters only split customers into new and long-standing, which is too coarse to act on. At four clusters, churn ranges from 4.8% to 48.7%, which separates customers worth contacting from customers who stay. Segments are named from the cluster centres (below or above the training average of 32.4 months and $64.54), not numbered, because cluster numbers can change between runs.

2.4 Predictive Model: Artificial Neural Network

A compact feed-forward neural network was built to estimate each customer's probability of churning. It takes the 10 prepared features as input. The customer segment is not an input: segments were combined with the model's predictions afterwards to interpret results.

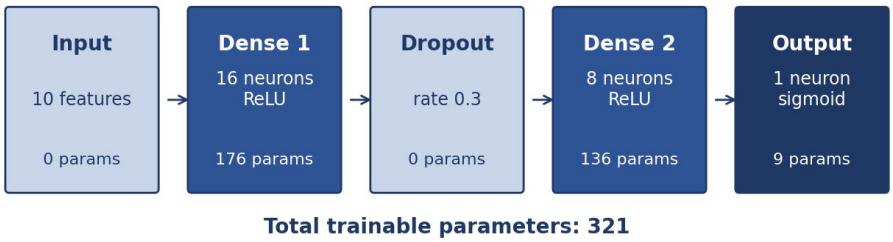


Figure 2: ANN architecture (321 trainable parameters)

Table 7: Layer-by-layer architecture

Layer	Type	Size	Activation	Parameters
Input	Input	10 features	None	0
Hidden 1	Dense	16 neurons	ReLU	176
Regularisation	Dropout	Rate 0.3	None	0
Hidden 2	Dense	8 neurons	ReLU	136
Output	Dense	1 neuron	Sigmoid (churn probability)	9
Total				321

The network is deliberately small. With 10 input features and 5,634 training rows, a larger network would almost certainly overfit. ReLU trains quickly and avoids vanishing gradients; dropout switches off 30% of neurons at each step to reduce overfitting; the sigmoid output gives a probability between 0 and 1, which suits a yes/no question.

Table 8: Training configuration

Training setting	Choice	Reason
Optimiser	Adam	Adapts the learning rate; works well for small networks
Loss function	Binary cross-entropy	Standard loss for a two-class problem
Batch size / epochs	32 / up to 50	Cap on training length
Early stopping	Validation loss, patience 10, best weights restored	Stops before overfitting
Class weights	0.68 for no-churn, 1.88 for churn	Churners count about 2.8 times more, so the model favours catching them
Training outcome	Stopped at epoch 33; best weights from epoch 23	Validation loss tracked training loss closely

The class weights are a deliberate trade-off. The model accepts more false alarms in exchange for catching more real churners, which is what a retention team wants. This is also why accuracy is not the main yardstick.

2.5 How the Model Was Evaluated

The model was scored on the 1,409-customer test set, which it never saw during training. Four measures are reported, because each answers a different business question:

- **Recall:** of the customers who really churned, what share did the model catch?
- **Precision:** of the customers the model flagged, what share really churned?
- **Accuracy:** what share of all predictions were correct?
- **ROC-AUC:** how well does the model separate churners from non-churners across all possible cut-offs? 0.5 is guessing; 1.0 is perfect.

Permutation importance then shuffled each feature in turn and measured how much performance fell, showing which features the model really relies on.

Part A: Summary of Key Findings

Assignment brief reference: 3-2, item 1, Summary of key findings from customer segmentation and churn prediction analysis.

A.1 Headline Findings

1. Overall, 1,869 of 7,043 customers (26.54%) have churned.
2. Churn is concentrated. New high spenders are about 31.5% of training customers but produce 57.8% of training-set churners (61% of actual churners on the test set).
3. Short tenure plus a high bill is the risk pattern. The riskiest segment averages 14.9 months and $81.10 a month, and 55.7% of its customers are on month-to-month contracts.
4. Long-term low spenders are the safest group, churning at about 5%.
5. The ANN catches 73.8% of churners, with an ROC-AUC of 0.808, so it ranks customers by risk well even though precision is modest.
6. The model relies mainly on tenure (0.21) and monthly charges (0.11), which are the same two features that separate the segments.

A.2 Customer Segmentation Results

Table 9: Customer segments and churn rates

Segment	Customers (training)	Share	Avg tenure	Avg monthly charge	Churn (training)	Churn (test)
New high spenders	1,775	31.5%	14.9 months	$81.10	48.7%	50.6%
New low spenders	1,381	24.5%	10.5 months	$32.10	about 24.8%*	22.9%
Long-term high spenders	1,512	26.8%	58.7 months	$93.80	16.1%	14.2%
Long-term low spenders	966	17.1%	54.5 months	$34.80	4.8%	5.2%

** Derived: the training set had 1,495 churners; subtracting the other three segments (864, 243 and 46) leaves 342 for new low spenders, or 24.8% of 1,381. Training figures come from the clustering analysis; test figures come from the model analysis.*

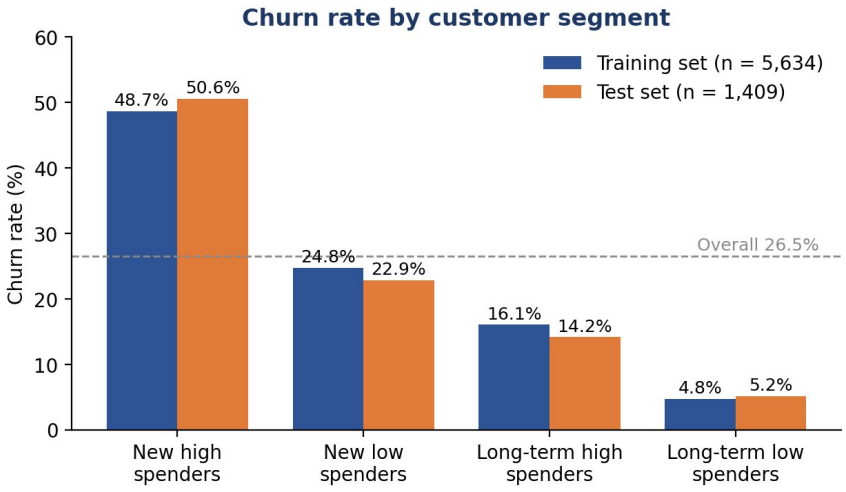


Figure 3: Churn rate by segment on the training and test sets

The pattern holds on data the clustering never saw. New high spenders churn at 48.7% in training and 50.6% on the test set, and long-term low spenders at 4.8% and 5.2%. The ten-fold gap between the best and worst segments is therefore a stable feature of the customer base, not a quirk of one sample.

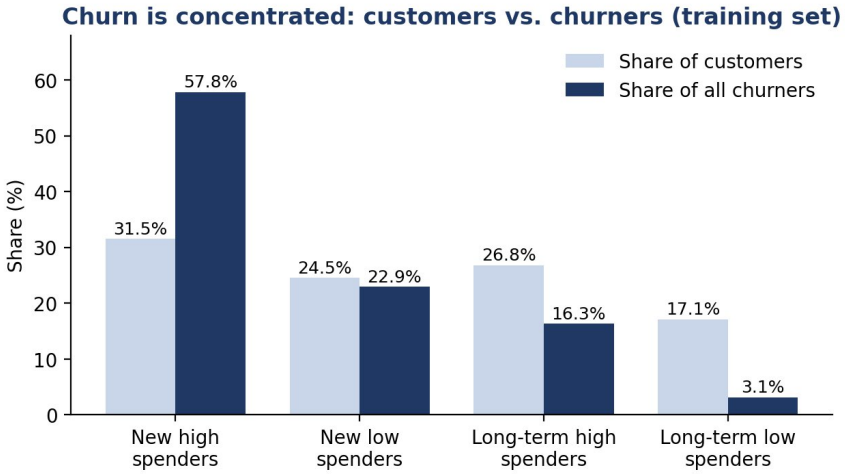


Figure 4: Share of customers compared with share of churners, by segment (training set)

Multiplying each segment's size by its churn rate gives the churners it produces. New high spenders produced 864 of the 1,495 training-set churners (57.8%), new low spenders about 342 (22.9%), long-term high spenders 243 (16.3%) and long-term low spenders 46 (3.1%). One segment in four therefore carries well over half of all churn.

A.3 Churn Prediction Results

Table 10: ANN performance on the unseen test set

Measure	Result	Meaning
Accuracy	72.3%	1,019 of 1,409 customers classified correctly
Recall (churners)	73.8%	276 of the 374 real churners were caught
Precision (churners)	48.6%	276 of the 568 customers flagged actually churned
ROC-AUC	0.808	Good separation of churners from non-churners
F1-score (derived)	0.586	Balance of precision and recall
Specificity (derived)	71.8%	743 of the 1,035 customers who stayed were correctly left alone

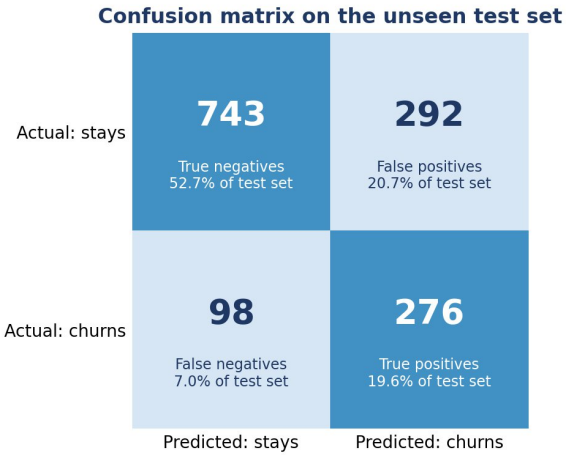


Figure 5: Confusion matrix for the test set (1,409 customers)

Reading the matrix: the model correctly flagged 276 churners and correctly cleared 743 customers who stayed. It missed 98 churners (false negatives) and raised 292 false alarms (false positives). Those 276 customers are the actionable group: people the retention team could contact before they leave.

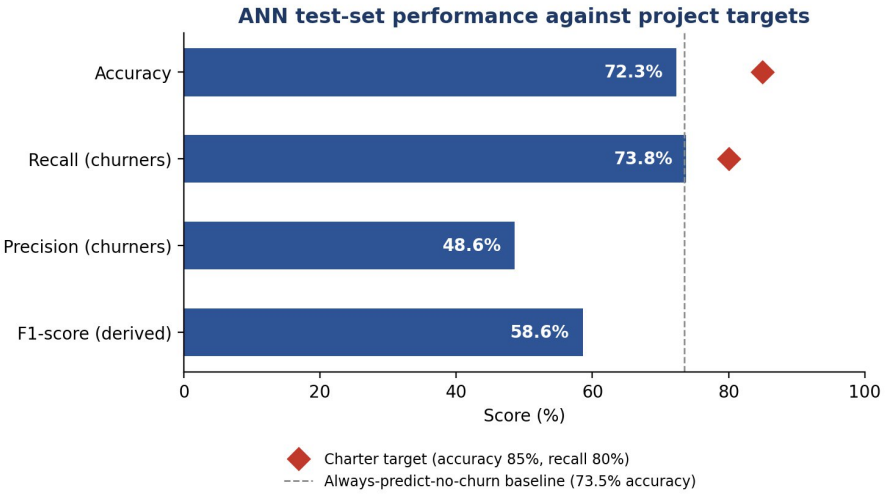


Figure 6: ANN results compared with the charter targets and the no-churn baseline

Two points of context matter. First, accuracy alone is misleading here, because always predicting “no churn” scores 73.5%; the model’s 72.3% is slightly below that, which is the expected cost of weighting the model towards catching churners. Second, the model flags about 40% of customers (568 of 1,409) to find 74% of churners. Its precision of 48.6% is about 1.8 times the base churn rate of 26.5%, so contacting a flagged customer is substantially better than contacting one at random.

A.4 Putting Segmentation and Prediction Together

Combining the model’s results with the segments tells a single story. The segment with the highest churn, new high spenders, is also the one the model relies on tenure and monthly charges to find: it is the short-tenure, high-bill group. On the test set it churns at 50.6% and accounts for 61% of all actual churners while making up only about a third of customers. That is the most important actionable finding of the project.

> **Key takeaway**
> Churn is not a broad problem across the customer base. It is concentrated in one identifiable group of new customers on high bills, and the model can flag individual customers in that group before they leave.

Part B: Factors Contributing to Churn and Retention

Assignment brief reference: 3-2, item 2, Identification and analysis of factors contributing to churn and retention, based on the ANN model's predictions.

B.1 Approach

Three independent views of the data were used and then compared, so that no single method carries the conclusions: a descriptive churn driver analysis (contract type, monthly charges and tenure), the K-Means segment profiles, and permutation importance on the trained ANN. A factor is treated as a confirmed driver when the views agree.

B.2 What the Neural Network Relies On

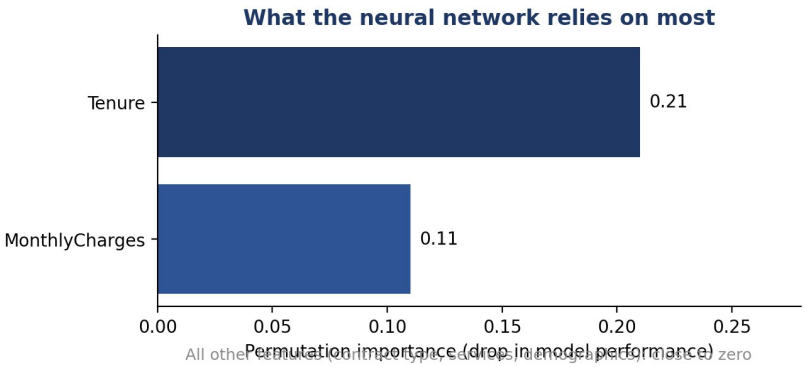


Figure 7: Permutation importance of the ANN's inputs

Permutation importance measures how much the model's performance drops when one feature is shuffled. Tenure is by far the strongest, at 0.21, roughly twice any other feature. Monthly charges is second, at 0.11. Everything else, including contract type, services and demographics, contributes almost nothing. In plain terms: newer customers on higher bills are the customers most likely to churn.

B.3 The Factors in Detail

Tenure

Tenure is the strongest driver and also separates the segments. The two new-customer segments average 14.9 and 10.5 months; the two long-term segments average 58.7 and 54.5. Customers early in the relationship have not built up habits, bundled services or loyalty, so they are the easiest for a competitor to win. Both long-term segments churn far less than the new high spenders.

Monthly charges

Monthly charges is the second driver, and it interacts with tenure. A high bill alone does not drive customers away: long-term high spenders pay the most ($93.80) yet churn at only 16.1% in training and 14.2% in testing. A high bill combined with short tenure is what produces roughly 50% churn. New customers paying a lot may not yet feel the value matches the price.

Contract type

The descriptive analysis identified contract type as a driver, and the segment profile supports it: 55.7% of new high spenders are on month-to-month plans, so nothing holds them. However, the neural network assigns contract type very little importance on its own. A likely explanation, which should be tested, is that

contract type overlaps with tenure and charges (month-to-month customers tend to be newer), so the model gets that information from the two stronger features. For retention, contract type remains the most practical lever, because it is the one the company can change; it is a lever more than a predictor.

Other features

Services subscribed and customer demographics contribute almost nothing to the model's predictions. Internet service type was noted as a secondary factor in the descriptive analysis but does not stand out in the model. We therefore do not recommend demographic targeting.

B.4 Evidence Compared

Table 11: Evidence for each factor across the three analyses

Factor	Churn driver analysis	Segment profiles	ANN importance	Conclusion
Tenure	Key driver	Splits new from long-term	0.21 (strongest)	Confirmed primary driver
Monthly charges	Key driver	Splits high from low spenders	0.11 (second)	Confirmed primary driver
Contract type	Key driver	55.7% month-to-month in riskiest segment	Near zero	Practical lever; overlaps with tenure
Internet service	Secondary	Does not define segments	Near zero	Minor
Demographics	Not a driver	Not used to split segments	Near zero	Not recommended for targeting

B.5 Factors Associated with Retention

The same evidence shows what keeps customers. Long tenure is the clearest retention factor: both long-term segments churn well below the 26.5% average. Lower bills also help, because long-term low spenders are the most loyal group at about 5% churn. Fixed commitment appears to matter too, since the segment most exposed to month-to-month plans is the one that leaves most. Customers who are past their first year or so, and who are comfortable with their bill, are the least likely to leave.

B.6 The Revenue Lens

Counting customers is not the same as counting money. If churners are assumed to pay their segment's average bill, the monthly billing associated with churners in the training set is as follows.

Table 12: Indicative monthly billing of churners (training set; assumes segment-average bills; new low spender count derived)

Segment	Churners (training)	Avg bill	Monthly billing of churners	Share of billing lost
New high spenders	864	$81.10	about $70,070	66.5%
New low spenders	about 342	$32.10	about $10,980	10.4%
Long-term high spenders	243	$93.80	about $22,790	21.6%
Long-term low spenders	46	$34.80	about $1,600	1.5%

New high spenders account for about two thirds of the billing lost, a larger share than their share of churners. Long-term high spenders are a smaller group of churners but about a fifth of the billing lost, because each customer is worth so much.

Part C: Recommendations for Targeted Retention Strategies

Assignment brief reference: 3-2, item 3, Recommendations for implementing targeted retention strategies.

C.1 Principles

- Target by segment. A single campaign for everyone would spend money on customers who are not at risk and under-invest in the group that matters.
- Match the action to the reason for leaving. New high spenders lack commitment; long-term high spenders carry high value; long-term low spenders are already loyal.
- Spend in proportion to risk and value. Retention budget follows both the churn rate and the billing at stake.
- We recommend; the company implements. Implementation is outside this project's scope.

C.2 Priority 1: New High Spenders, Convert to Fixed-Term Contracts

Why. This segment is about a third of customers, produces 57.8% of training churners (61% on the test set) and carries about 66.5% of the billing lost. Churn is about 49% to 51%, and 55.7% are on month-to-month plans.

- **Action:** offer a fixed-term contract (12 to 15 months) priced at or below the $81.10 these customers already pay, offered within their first 15 months.
- **Targeting:** use the ANN's churn probability to contact the highest-risk customers in this segment first.
- **Why it should work:** a commitment removes the easy exit, and the price anchor avoids giving them a new reason to shop around.
- **Expected effect:** the largest single reduction in churn available to the business, because it acts on more than half of all churn.

C.3 Priority 2: Long-Term High Spenders, Protect the Relationship

Why. Only 243 of 1,512 left (16.1%), but at $93.80 a month that is about $22,800 in monthly billing, around a fifth of all billing lost.

- **Action:** priority customer support, tenure-based rewards and recognition, and proactive check-ins before contract renewal.
- **Why not discounts:** these customers are not leaving over commitment or price sensitivity in the way new customers are. A discount would cut margin on loyal, high-value customers.
- **Expected effect:** protects the highest revenue per customer.

C.4 Priority 3: Long-Term Low Spenders, Upsell Instead of Retain

Why. Only 46 of 966 left (4.8%), the lowest churn of any segment, so retention spend here earns the least.

- **Action:** spend nothing extra on retention. Use this group as an upgrade list instead.
- **Opportunity:** they pay $34.80 a month against $93.80 for long-term high spenders with almost the same tenure, which suggests room to grow revenue from a loyal base.
- **Note:** this recommendation generates revenue; it does not prevent loss. It should be judged separately from the churn goal.

C.5 Monitor: New Low Spenders

New low spenders (about a quarter of customers, churn 22.9% to 24.8%) have the new-customer risk profile but low bills, so the billing at risk is small (about 10% of the total). We recommend standard onboarding and engagement, and monitoring this segment in reporting, rather than a costly dedicated campaign.

C.6 Use the Model as an Early-Warning Tool

The ANN gives each customer a churn probability. Used well, it moves the company from reactive to proactive retention.

- Score customers on a regular schedule (for example monthly) and send the highest-risk new high spenders to the retention team.
- Pick the cut-off to suit the cost of the offer. A low-cost offer can use a lower cut-off and accept more false alarms; an expensive one should use a higher cut-off.
- Remember that about half of flagged customers would have stayed (precision 48.6%), so offers should be cheap or conditional.

C.7 Recommendations at a Glance

Table 13: Summary of recommendations

Priority	Segment	Recommended action	Main reason	Suggested owner
1	New high spenders	Fixed-term contract offer, 12 to 15 months, at or below $81.10	57.8% to 61% of churners; no commitment	Sales and retention
2	Long-term high spenders	Priority support and tenure rewards, not discounts	About a fifth of billing lost	Customer care
3	Long-term low spenders	No retention spend; upgrade campaign	Lowest churn (about 5%); billing gap of $59	Marketing
Monitor	New low spenders	Standard onboarding; track in reports	Small billing at risk	Customer care

C.8 Suggested Implementation Roadmap

Implementation is for the company to decide and run. A sensible order, for the company's consideration, is:

Table 14: Proposed roadmap (for the company's consideration)

Phase	Timing	Activities
1. Validate and prepare	Weeks 1 to 4	Confirm segment definitions on current data; agree offer costs and cut-offs; set up scoring of customers with the ANN
2. Pilot	Weeks 5 to 12	Run the fixed-term offer with a sample of high-risk new high spenders against a comparison group; start the loyalty programme for long-term high spenders
3. Roll out	Months 4 to 6	Extend what worked; launch the upgrade campaign for long-term low spenders
4. Review	Ongoing	Track the measures below, retrain the model when performance drifts

C.9 Suggested Measures of Success

Table 15: Suggested measures (targets to be set by the company)

Measure	Why it matters
Churn rate of new high spenders	Direct test of Priority 1
Share of new high spenders moving to fixed-term contracts	Shows whether the offer is taken up
Churn rate of pilot group against comparison group	Isolates the effect of the offer
Monthly billing retained from long-term high spenders	Tests Priority 2 in money terms
Upgrade take-up among long-term low spenders	Tests Priority 3
Model recall and precision on new data	Shows when the ANN needs retraining

C.10 Costs, Benefits and Risks

This project did not include detailed financial modelling, so no return on investment is claimed. The company can judge each action with a simple test: contacting a customer is worthwhile when the chance they would leave, multiplied by the chance the offer saves them, multiplied by the value of keeping them, is greater than the cost of the offer. The segment and billing figures in this report supply the first and third inputs.

- **Discount leakage:** an offer given to customers who would have stayed costs money. Mitigation: use the churn probability to target, and use conditional offers.
- **Model error:** the model misses about one in four churners. Mitigation: pair it with simple rules, such as flagging all customers in their first year on high bills.
- **Changing behaviour:** patterns may shift. Mitigation: refresh the analysis regularly.

Part D: Limitations and Proposed Solutions

Assignment brief reference: 3-2, item 4, Documentation of limitations and challenges, with proposed solutions or mitigations.

D.1 The Model Did Not Reach the Charter Targets

This is the most important limitation. The charter set a minimum of 85% accuracy and 80% recall. The model achieved 72.3% and 73.8%. Recall is within about six points of its target; accuracy is more than twelve points short.

Context explains part of the gap without excusing it. First, the model was weighted to favour catching churners, which lowers accuracy by design. Second, the data has only 10 input features and a modest 5,634 training rows. Third, the model leans on two features, which suggests the dataset contains limited extra signal, such as customer service contacts or usage patterns. An 85% accuracy target may therefore be unrealistic with this data, but the team did not test that claim, and the shortfall should be reported as it stands.

D.2 Limitations and How to Address Them

Table 16: Limitations, their effect and proposed solutions

Limitation	Effect on the findings	Proposed solution
Model below charter targets (accuracy 72.3% vs 85%; recall 73.8% vs 80%)	The model misses about one in four churners and its accuracy is below the no-churn baseline	Tune the decision cut-off to raise recall; run a wider hyperparameter search; add features; benchmark against other models such as gradient boosting or balanced random forests
Low precision (48.6%)	About half of flagged customers would have stayed; 292 false alarms in the test set	Use low-cost or conditional offers; choose a higher cut-off for expensive offers; consider resampling methods such as SMOTE
Few informative features	Model relies on tenure and charges; others add almost nothing	Collect richer data (service calls, complaints, usage, payment behaviour) in a future phase
Weak clustering (silhouette 0.203)	Segments sit close together; they are action lists, not proof of four distinct customer types	Use segments to prioritise; re-test segmentation as data changes; compare with other methods
Segments driven by tenure and charge	Other features are 0 or 1 flags and so have less pull on distance	Keep contract and internet type as separate predictors; consider feature weighting
Contract type shows low model importance	Its effect may be hidden by overlap with tenure	Test contract type in a controlled pilot; check correlation with tenure
Single train and test split	Results come from one split of the data	Use cross-validation to give a range for each measure
Static, historical data only	Behaviour and market conditions may change	Refresh the analysis regularly with new data
Model not deployed or monitored	No ongoing tracking of results	Plan deployment and monitoring if the company proceeds
302 duplicate rows kept	May be distinct customers, or rarely accidental duplicates	Re-check against a unique customer ID if one becomes available
No financial modelling	Costs and returns of each strategy are not quantified	Run a cost-benefit assessment before large-scale rollout

D.3 Project Challenges and How They Were Handled

- **Scaling leakage.** The scaler was first fitted on all rows. Rohan found this and fixed it by fitting on the training set only, and documented the proof: training columns centre at exactly zero while test columns come out close but not equal.
- **Compressed end of Stage 2.** Two clustering tickets closed after their due dates, and upload and video work fell on the same day. The team set a noon cut-off for files, a repository check by 3 pm and recording by 6 pm, so both submissions went in before the deadline.
- **Silhouette score disagreeing with the elbow.** The team chose four clusters on business usefulness and showed the evidence for the choice, rather than hiding the disagreement.
- **Dependencies between roles.** Findings, recommendations and the video depend on the model results, so the team agreed dates in advance, asked members to flag at-risk dates the same day, and planned the Stage 3 submission a day early.

D.4 Proposed Next Iteration

1. Tune the decision cut-off and report recall and precision at several cut-offs, so the company can pick the trade-off it prefers.
2. Run cross-validation and a hyperparameter search for the ANN, and benchmark it against at least one tree-based model.
3. Add richer features from customer service and usage data, if the company can provide them.
4. Pilot the Priority 1 recommendation with a comparison group, and use the result to test contract type directly.
5. Cost each recommendation and decide where the model should be deployed and how it will be monitored.

3. Video Demonstration

Assignment brief reference: 3-2, item 5, Video demonstration (10 to 15 minutes).

The team recorded a single video of 10 to 15 minutes demonstrating the development and evaluation of the ANN model. It follows the order below and uses the figures in this report.

Table 17: Video plan

Presenter	Topic	Content
Shradha Malla	The model	Data overview, the ANN architecture, training setup and early stopping, test-set metrics and confusion matrix, permutation importance, churn by segment
Himal Kunwar	Findings, recommendations, limitations	Segment results and the concentration of churn, factors behind churn, the three recommendations, limitations and solutions
Rohan Sharma Kharel	Model inputs	How the prepared data and saved scaler feed the network
Ramesh Kunwar	Segments and the model	How the four segments were produced and how they are combined with predictions

The video covers every element the brief requires: the defined architecture, model training, evaluation of performance, key findings and insights with visualisations of churn predictions and contributing factors, the recommendations, and the limitations and challenges. The video file is submitted by the Project Manager together with the repository link and this report.

4. Alignment with Stage 3 Requirements

The rubric awards full marks only when every element of the brief is present. The table maps each element to where it is addressed.

Table 18: Requirements checklist

Brief requirement	Where addressed	Deliverable
3-1.1 Defined architecture of the ANN	Section 2.4; ANN_Architecture.pdf in Predictive_Modeling	Predictive_Modeling folder
3-1.2 Trained ANN on the dataset	Section 2.4; trained model file and training code in Predictive_Modeling	Predictive_Modeling folder
3-1.3 Predictions and performance evaluation	Section 2.5 and Part A.3; metrics, confusion matrix and predictions in Predictive_Modeling	Predictive_Modeling folder
3-2.1 Summary of key findings	Part A	Final_Report
3-2.2 Factors contributing to churn and retention	Part B	Final_Report
3-2.3 Retention recommendations	Part C	Final_Report
3-2.4 Limitations and proposed solutions	Part D	Final_Report
3-2.5 Video demonstration (10 to 15 minutes)	Section 3	Video file
Repository link as the primary submission	Appendix C	Repository
Single submission by the Project Manager	Document information; Section 3	Canvas

5. Conclusion

This project set out to understand why customers leave a telecommunications company and to turn that into actions the business can take. The answer is specific. Churn is concentrated in new customers on high bills: new high spenders are about a third of customers, yet they produce more than half of all churners and about two thirds of the billing lost. Tenure and monthly charges are the two factors that matter most, in the clustering and in the neural network alike.

The neural network can flag at-risk customers, finding about three in four churners on unseen data, but it falls short of the accuracy and recall targets in the charter, and about half of the customers it flags would have stayed. We state this plainly and set out how to improve it. The recommendations do not depend on the model being perfect: they rest on a pattern that appears in the segmentation, in the model and on the test set.

We recommend that the company begin with a fixed-term offer for new high spenders, protect its high-value long-term customers with service rather than discounts, and treat its most loyal low spenders as an upgrade

opportunity. A small pilot with a comparison group is the sensible first step, and would also answer the questions this analysis could not.

Acknowledgements. The team thanks our mentor, Atin Chaudhary, for guidance and feedback across all three stages.

Appendices

Appendix A: Glossary

Term	Meaning
Churn	A customer ending their service with the company
Segment / cluster	A group of customers who are similar to each other, found with K-Means
K-Means	A method that groups customers around a set number of centres
Elbow method	Choosing the number of clusters where extra clusters stop helping much
Silhouette score	How well separated clusters are; higher is better, 1 is perfect
ANN	Artificial Neural Network, a model made of layers of connected units
ReLU / Sigmoid	Two activation functions; sigmoid outputs a probability between 0 and 1
Dropout	Randomly switching off units while training to reduce overfitting
Early stopping	Stopping training when validation performance stops improving
Class weights	Making the model count the rarer class (churners) more heavily
Recall	Share of real churners that the model caught
Precision	Share of flagged customers who really churned
ROC-AUC	How well the model ranks churners above non-churners; 0.5 is guessing
Permutation importance	How much performance falls when one feature is shuffled
Data leakage	Test information influencing training, which inflates results

Appendix B: Metric Calculations

Confusion matrix, test set: true negatives 743, false positives 292, false negatives 98, true positives 276 (total 1,409).

Table 19: Metric calculations (F1, specificity, flag rate and lift derived from the matrix)

Measure	Formula	Calculation	Result
Accuracy	(TP + TN) / total	(276 + 743) / 1,409	72.3%
Recall	TP / (TP + FN)	276 / (276 + 98) = 276 / 374	73.8%
Precision	TP / (TP + FP)	276 / (276 + 292) = 276 / 568	48.6%
F1-score	2 × P × R / (P + R)	2 × 0.486 × 0.738 / (0.486 + 0.738)	0.586
Specificity	TN / (TN + FP)	743 / (743 + 292) = 743 / 1,035	71.8%
Flag rate	(TP + FP) / total	568 / 1,409	40.3%
Lift over base rate	Precision / base churn rate	0.486 / 0.2654	about 1.8

Appendix C: Repository Structure

Folder	Contents
Data	Raw dataset, unchanged

Folder	Contents
Data_Preparation	Preparation notebook and script, preprocessed dataset, train and test sets, encoding map, fitted scaler, scaling techniques document
Clustering_Analysis	Elbow method outputs, selection metrics, trained K-Means model, cluster assignments, five labelled visualisations, README with segment profiles
Predictive_Modeling	ANN_Architecture.pdf, Stage3_ANN_Churn.ipynb, trained model file, training_history.png, metrics.csv, confusion_matrix.png, permutation_importance.csv, churn_by_segment.png, predictions
Final_Report	This report

Appendix D: Task Register

Ticket	Task	Owner
WILDA4-43	Training and testing split	Rohan Sharma Kharel
WILDA4-44	Optimal number of clusters (elbow method)	Ramesh Kunwar
WILDA4-45	Train the K-Means model	Ramesh Kunwar
WILDA4-46	Visualise and label clusters	Ramesh Kunwar
WILDA4-47 / 81	Define ANN architecture; architecture document	Shradha Malla
WILDA4-48	Train the ANN and tune convergence	Shradha Malla
WILDA4-49	Evaluate performance and analyse predictions	Shradha Malla
WILDA4-50	Findings summary	Himal Kunwar
WILDA4-51	Retention recommendations	Himal Kunwar
WILDA4-52	Limitations section	Himal Kunwar
WILDA4-58 / 80	Churn driver analysis; match drivers to segments	Himal Kunwar
WILDA4-79	Hand over cluster assignments	Ramesh Kunwar
WILDA4-87	Customer segmentation section	Ramesh Kunwar
WILDA4-93	Check model inputs against encoding map and scaler	Rohan Sharma Kharel
WILDA4-95	Churn and retention factors section	Rohan Sharma Kharel
WILDA4-88	Stage 3 video demonstration	Whole team
WILDA4-68 / 78	Repository and project management epic	Bhawana K C